M2795.007800
Guidance, Navigation, and Control System
for Reusable Launch Vehicles

Module 5.
Navigation and State Estimation



Contents

- Navigation Fundamentals & Sensor Modeling
  - Strapdown inertial navigation and navigation frames
  - Gyros, accelerometers, GNSS, radar altimeter, landing sensors
  - Sensor error models, calibration, and initial alignment
- State Estimation & EKF
  - Nominal INS + error-state EKF architecture
  - Prediction, covariance propagation, and measurement updates
  - Multi-rate fusion, dropout/outlier handling, and filter tuning
  - Verification, Monte Carlo testing, and Project 2 deliverables

Navigation converts imperfect sensor measurements into the state information used by guidance and control

Navigation Fundamentals & Sensor Modeling

Navigation: Bridge from Sensors to G&C



- Navigation answers three questions in real time

Where am I?	Position / altitude
How am I moving?	Velocity / angular rate
How am I oriented?	Attitude

For a reusable booster, navigation must r emain useful across:

- high-dynamic ascent
- stage separation and reorientation
- atmospheric reentry
- powered descent and precision landing

The same vehicle may need different sensor combinations in different phases

Navigation Requirements Change

Ascent
Attitude / Rate / Velocity

→

Separation / Return
Position / Velocity / Attitude

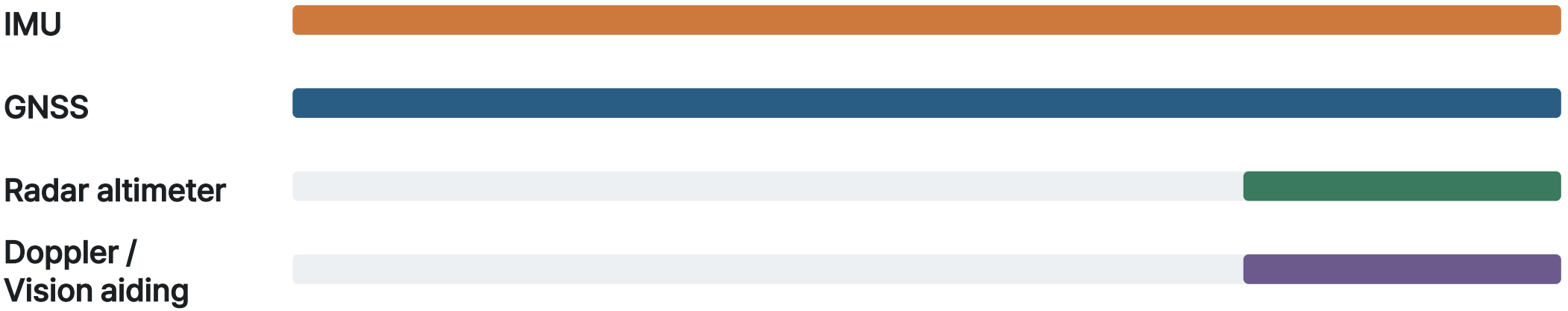
→

Atmospheric Entry
Energy state /
Attitude / Position

→

Terminal Landing
AGL altitude /
Vertical velocity / Lateral error

- Representative sensor availability / usefulness

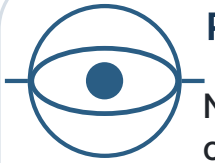


Design implication: do not assume one sensor provides all required states with the same quality throughout the mission

Methods of Navigation

- How can a vehicle determine where it is and how it is moving?

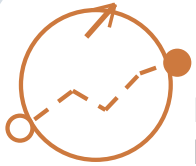
From observation → self-contained sensing → satellite positioning



Piloting

Navigate by direct visual observation of landmarks and vehicle motion

Visual / human



Dead-Reckoning

Estimate position from a known start using heading, speed/velocity, and elapsed time

Errors accumulate



Celestial Navigation

Determine position or attitude from observations of the Sun, Moon, or stars

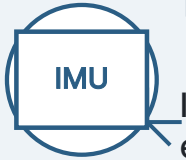
External natural references



Radio Navigation

Use radio beacons, radar, or tracking stations to determine range, bearing, or position

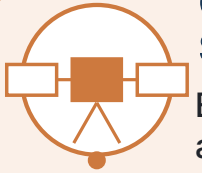
External radio reference



Inertial Navigation System (INS)

Integrate gyroscope and acc-elerometer measurements to compute attitude, velocity, and position

Self-contained • high-rate • drifts



Global Navigation Satellite System (GNSS)

Estimate absolute position and velocity from signals transmitted by navigation satellites.

Absolute reference • signal-dependent

Launch-vehicle perspective

Modern launch vehicles rely primarily on inertial navigation, often aided by GNSS and mission-specific external sensors

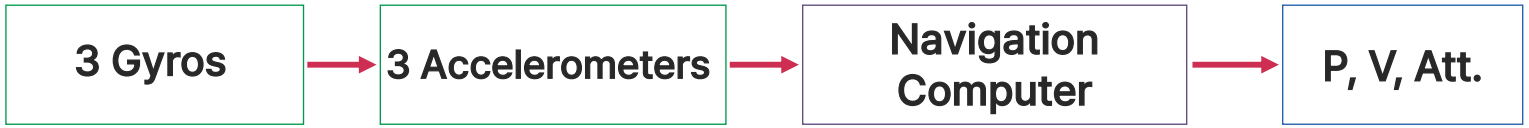
Launch Vehicle Navigation System Example



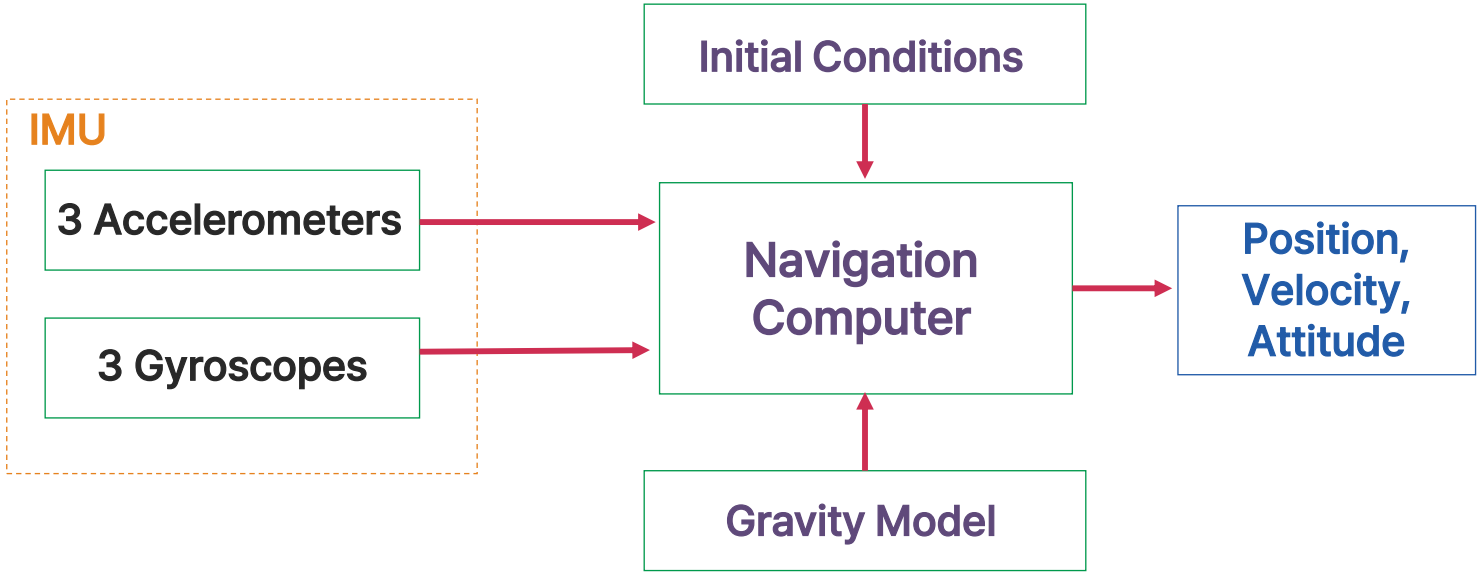
- Thales Avionics Quasar 3000
  - Overview
    - Strapdown inertial navigation unit used on Ariane 5
    - Based on three gyroscopes and three accelerometers
    - Provides attitude, velocity, and inertial reference data for launch vehicle guidance, navigation, and control.
    - Representative performance data
      - Attitude drift < 0.03 deg/h
      - Position drift < 1.5 km/h



Typical architecture



Elements of Inertial Navigation



IMU (Inertial Sensors)

Gyroscope: measurements of angular rate
Accelerometer: measurement of specific force (non-gravitational acceleration)

$$\vec{a} = \frac{\vec{F}}{m} + \vec{g}$$

Navigation Computation

Integration from the initial conditions
Attitude computation by integrating the gyro measurements
Velocity and position computation by integrating the accelerometer measurements together with the gravity model

INS vs. GNSS: Complementary, Not Competing

Characteristic	INS	GNSS
Principle	Dead reckoning	Range-based positioning
Short-term	Excellent	Noisy / lower rate
Long-term	Drifts with time	Absolute reference
Attitude	Available	Typically not direct
Self-contained	Yes	No
Output rate	High	Lower

INS strength

High-rate, continuous state propagation through aggressive vehicle motion

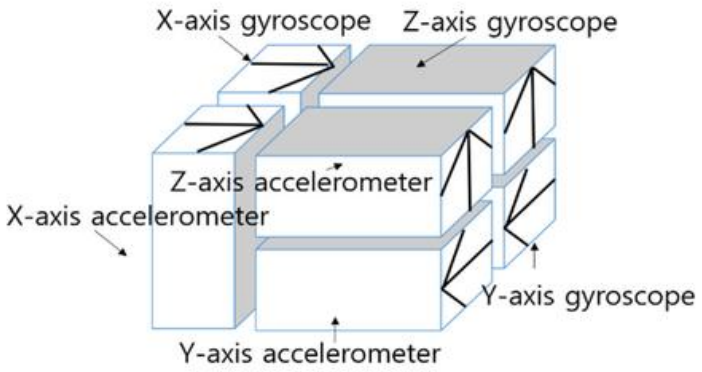
GNSS strength

Absolute position/velocity information prevents unbounded inertial drift when measurements are available

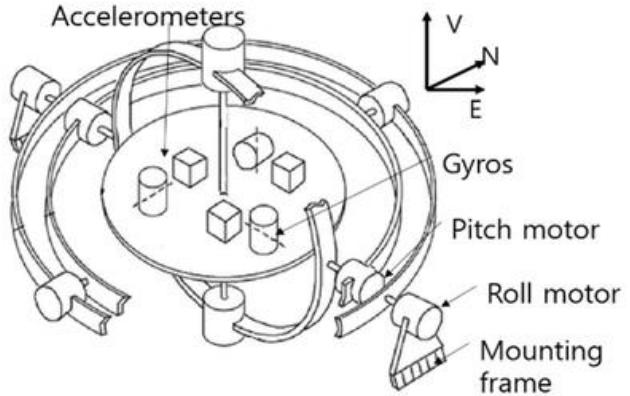
Project 2 design: propagate with IMU, correct with GNSS, and add landing-specific aiding near touchdown

Types of Inertial Navigation System

Strapdown INS



Gimbaled INS



Feature	Strap-down Method	Platform Method (Gimbaled INS)
Sensor Mounting	Bolted directly to the vehicle body	On a stabilized, gimbaled platform
Physical Complexity	Low (no moving parts, just a solid-state sensor cluster)	High (requires complex mechanical gimbals and motors)
Size, Weight, & Cost	Small, lightweight, and low-cost	Large, heavy, and very expensive
Computing Power Required	Very High (requires constant, high-speed coordinate transformations)	Low (the hardware does the stabilization work)
Reliability & Durability	Higher (shock-resistant, highly durable solid-state electronics)	Lower (mechanical joints and motors can wear out)
Sensor Dynamic Range	Higher (sensors must tolerate the full angular speed of the vehicle)	Lower (sensors don't experience intense vehicle rotations)

Launch-Vehicle Inertial Sensor Suite

Gyroscopes

Measure body angular rate

- RLG: ring laser gyro
- FOG: fiber-optic gyro
- MEMS / resonant gyros

Accelerometers

Measure specific force

- servo / pendulous
- vibrating beam
- MEMS

Why strapdown?

- Sensors rigidly mounted to the vehicle
- No mechanical stabilized platform
- Body rotation handled by software frame transform-ation
- Compact, robust, and well suited to launch vehicles

Strap-down Method	Platform Method (Gimbaled INS)
Delta II, Taurus, Pegasus, PSLV, N-II, KSLV-I, Atlas 5, Delta 4, Minotaur, Ariane 5, Vega, H-II/IIA, SLS, M-V, Falcon 1/9, CZ-8, KSLV-II	Saturn V, Atlas-Centaur, Ariane 1–4, Zenit, Proton, H-I, CZ-4, Space Shuttle

Historical launch-vehicle examples span gimbaled and strapdown systems; modern systems predominantly use s trapdown INS

Real Launch-Vehicle Example: KSLV-II INGU

Navigation / alignment performance (3σ)

Parameter	Specification
Azimuth alignment	0.025 deg
Level alignment	0.014 deg
Attitude @ 900 s	0.1 deg
Velocity @ 900 s	3.0 m/s
Position @ 900 s	1.5 km

Sensor configuration

3 FOGs + 4 accelerometers; strapdown inertial navigation

Parameter		Specification (3σ)
Alignment	Azimuth (optical alignment)	0.025 deg
	Level	0.014 deg
Navigation (900 sec)	Attitude	0.1 deg
	Velocity	3.0 m/s
	Position	1.5 km

Engineering lesson

Navigation requirements must be translated into sensor range, bias, drift, scale-factor, alignment, and computing specifications

Strapdown INS Mechanization



$$\dot{q} = \tfrac{1}{2}\,\Omega(\omega)\,q$$

$$\dot{v} = C_b^n f^b + g^n + \text{frame terms}$$

$$\dot{r} = v$$

Three recurring operations: ① integrate body rotation
② rotate measured specific force into the navigation frame
③ integrate acceleration to velocity and position

Reference Frames: Choose One and Be Consistent

ECI	Orbital / inertial dynamics	Inertial frame; no Earth rotation in position frame
ECEF	Ground-relative launch & landing	Fixed to rotating Earth; natural for GNSS / landing sites
Local NED / ENU	Terminal guidance & landing	Intuitive local position, velocity, altitude
Body	IMU & actuators	Gyro/accelerometer measurements, thrust, moments

Project 2 recommendation

Propagate position/velocity in ECEF (or a clearly defined local navigation frame), while IMU measurements remain in the body frame

Specific Force and Attitude Propagation

What an accelerometer measures

$$f^b = a^b - g^b$$

Accelerometers do not directly measure “gravity acceleration.”

They measure non-gravitational specific force.

Gyro → attitude

$$\dot{q} = \tfrac{1}{2}\,\Omega(\omega_m - b_g)\,q$$

Integrate angular rate, then normalize quaternion.

The attitude determines how accelerometer data are rotated into the navigation frame.

Attitude → acceleration

$$a^n = C^{b\,n} f^b + g^n$$

An attitude error rotates the large gravity vector incorrectly and quickly contaminates velocity and position.

Gyro bias → attitude error → incorrect gravity projection → acceleration error → velocity error → position error

Sensor Error Model for Simulation

$$z_m = (I + S + M)z_{true} + b + n$$

A practical IMU model separates repeatable and stochastic errors

Bias b
constant / slowly varying offset

Misalignment M
cross-axis coupling

Scale factor S
output proportional to true input

Noise n
white noise + bias instability

Project 2: implement errors in layers

Level 1	white noise only
Level 2	constant bias + white noise
Level 3	bias random walk + scale factor
Advanced	misalignment / temperature effects

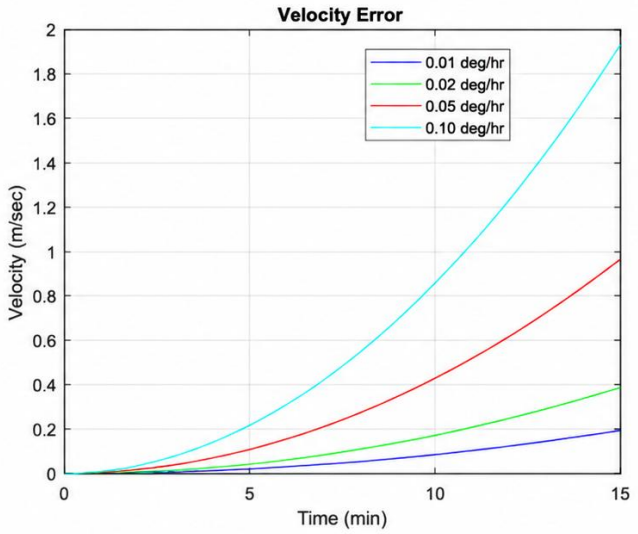
Why Tiny Gyro Bias Matters

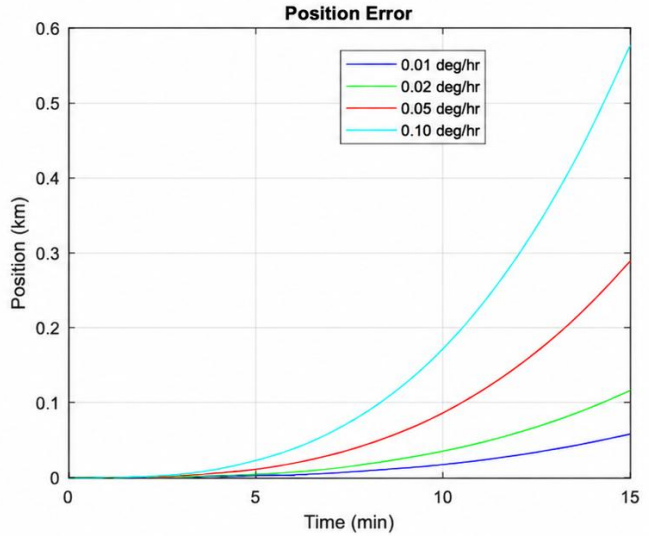
At approximately 1 g, a constant horizontal gyro bias produces rapidly growing navigation error

$\delta\theta \approx \delta\omega_b\, t$ — $\delta v \approx \frac{1}{2}\, g\, \delta\omega_b\, t^2$ — $\delta p \approx \frac{1}{6}\, g\, \delta\omega_b\, t^3$

attitude error | velocity error | position error

- **Horizontal Attitude error :** $\delta\theta = \int_0^t \delta\omega_B dt = \delta\omega_B t$
- **Horizontal Velocity error :** $\delta V_n = \int_0^t \delta a_N dt = \int_0^t g\delta\omega_B t dt = \frac{1}{2} g\delta\omega_B t^2$
- **Horizontal Position error :** $\delta p_n = \int_0^t \delta V_N dt = \int_0^t \frac{1}{2} g\delta\omega_B t^2 dt = \frac{1}{6} g\delta\omega_B t^3$





Calibration and Initial Alignment

Calibration
Estimate repeatable bias, scale factor, misalignment

→

Coarse alignment
Use gravity + Earth rotation / external azimuth reference

→

Fine alignment
Estimate small attitude errors and sensor biases

→

Launch
Start navigation with x_0 and P_0

Coarse alignment intuition

- accelerometer average → gravity direction
- gyro average → Earth-rate direction (with sufficient sensor quality)
- external azimuth / optical alignment may be used

Goal: determine the initial body-to-navigation attitude

Fine alignment / filter initialization

- refine small attitude errors
- estimate gyro / accelerometer bias
- set initial covariance P_0 realistically
- verify static velocity remains near zero

Poor alignment becomes a systematic navigation error after liftoff

Reusable Booster: Aiding Sensors by Flight Phase

Sensor / source	Ascent	Return / Entry	Terminal landing	Primary information
IMU	●	●	●	ω, specific force
GNSS	●	●	●*	absolute position / velocity
Radar altimeter			●	AGL altitude
Doppler radar / lidar			○	relative velocity / altitude
Vision / TRN		○	○	terrain-relative position
Vehicle models	●	●	●	mass, gravity, atmosphere

● baseline / highly useful ○ optional or mission-dependent * availability and quality may vary with geometry and environment

Reusable flight adds a precision-landing problem:
altitude and relative horizontal/vertical motion matter much more near touchdown.

Project 2 Sensor Models

IMU

200 Hz*

$$\omega_m = \omega + b_g + n_g$$
$$f_m = f + b_a + n_a$$

bias / noise / update timing

GNSS

10 Hz*

$$r_m = r + n_r$$
$$v_m = v + n_v$$

bias / noise / update timing

Radar Altimeter

20 Hz*

$$h_m = h_{AGL} + n_h$$

bias / noise / update timing

Implementation rule

- Generate measurements from x_true, never from x_hat
- Each sensor has its own rate and activation logic
- Store both truth and measurements so filter errors can be diagnosed

* Representative project rates — not specifications of a particular launch vehicle

Implementation Checklist

1	**Frames**	Define ECEF/NED/body conventions and rotation direction
2	**Strapdown INS**	Propagate attitude, velocity, and position from IMU
3	**Sensor model**	Add IMU bias/noise; GNSS position/velocity; terminal altitude sensor
4	**Initialization**	Specify initial state, alignment error, and sensor biases
5	**Verification**	Static test → simple motion → flight trajectory

Next: How to combine the dynamic model and aiding measurements to estimate the state optimally

State Estimation
& Extended Kalman Filter

TBD